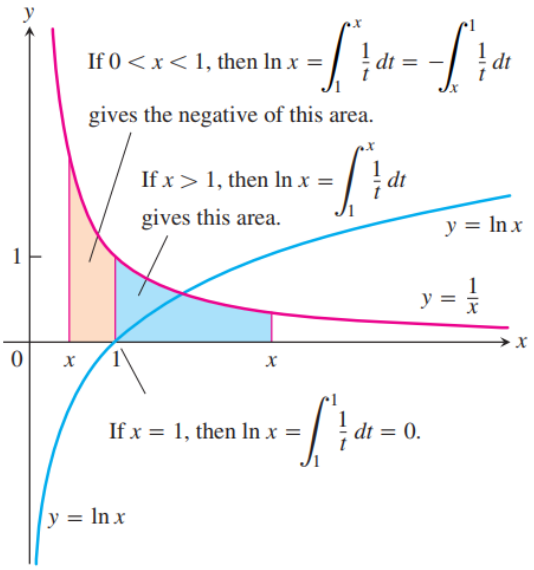
Natural Logarithms

The natural logarithm of a positive number x, written as $\ln x$, is the value of an integral.

Definition. *The Natural Logarithm Function*

$$\ln x = \int_1^x \frac{1}{t}\,dt, x > 0.$$



The graph of $y = \ln x$ and its relation to the function $y = \frac{1}{x}, x > 0$.

The graph of the logarithm rises above the x-axis as x moves from 1 to the right.

It falls below the axis as x moves from 1 to the left.

We also have

$$\ln 1 = \int_1^1 \frac{1}{t}\,dt = 0.$$

Definition. *The number e*

The number e is that number in the domain of the natural logarithm satisfying

$$\ln(e) = \int_1^e \frac{1}{t}\,dt = 1.$$

Geometrically, the number e corresponds to the point on the x-axis for which the area under the graph of and above the interval $[1, e]$ is the exact area of the unit square.

*The area of the region shaded blue in Figure above is 1 sq unit when $x = e$.

The Derivative of $y = \ln x$

By the first part of the Fundamental Theorem of Calculus,

$$\frac{d}{dx}\ln x = \frac{d}{dx}\int_1^x \frac{1}{t}\,dt = \frac{1}{x}.$$

** *Fundamental Theorem of Calculus will be discussed in Integration.*

For every positive value of x, we have

$$\frac{d}{dx}\ln x = \frac{1}{x}.$$

$\frac{d}{dx}\ln(f(x)) = \frac{1}{f(x)} f'(x)$

$\frac{d}{dx}\ln|x| = \begin{cases} \frac{1}{x}, & x>0 \\ \frac{1}{-x}\cdot(-1), & x<0 \end{cases} = \frac{1}{x}$

$\frac{d}{dx}\ln|x| = \frac{1}{x}, \quad x\neq 0.$

If u is a differentiable function x whose values are positive ~~so that $\ln u$ is defined~~, then applying the Chain Rule to the function $y = \ln u$ gives

$\int \frac{1}{x}dx = \ln|x| + C$

$$\frac{d}{dx}\ln u = \frac{1}{u}\cdot\frac{du}{dx}, \qquad u > 0.$$

Example. *Derivatives of Natural Logarithms*

(a) $\frac{d}{dx}\ln 2x = \frac{1}{2x}\frac{d}{dx}(2x) = \frac{1}{x}$

(b) $\frac{d}{dx}\ln(x^2+3) = \frac{1}{x^2+3}\cdot\frac{d}{dx}(x^2+3) = \frac{2x}{x^2+3}.$

(c) $\frac{d}{dx}(\ln x)^4 =$

(d) $\frac{d}{dx}(\ln|x|) =$

$\frac{d}{dx}\ln(2x) = \frac{1}{2x}\cdot\frac{d}{dx}(2x) = \frac{2}{2x} = \frac{1}{x}.$

$\frac{d}{dx}\left(\ln(x^2+3)\right) = \frac{1}{x^2+3}(2x) = \frac{2x}{x^2+3}$

$\frac{d}{dx}\left((\ln x)^4\right) = 4(\ln x)^3\cdot\frac{1}{x}$

$\frac{d}{dx}\left(\ln|x|\right) = \frac{1}{x}, \quad x\neq 0.$

Derivative of $\ln|x|$

$$\frac{d}{dx}\ln|x| = \frac{1}{x}, x \neq 0$$

Notice from Example (a) that the function $y = \ln 2x$ has the same derivative as the function $y = \ln x$. This is true of $y = \ln bx$ for any constant b, provided that $bx > 0$:

$$\frac{d}{dx}\ln bx = \frac{1}{bx}\frac{d}{dx}(bx) = \frac{1}{bx}(b) = \frac{1}{x}.$$

Properties of Logarithms

Logarithms, invented by John Napier, were the single most important improvement in arithmetic calculation before the modern electronic computer. The properties of logarithms reduce multiplication of positive numbers to addition of their logarithms, division of positive numbers to subtraction of their logarithms, and exponentiation of a number to multiplication of its logarithm by the exponent.

* $\ln(a+b) \neq \ln a + \ln b$

THEOREM 2: Algebraic Properties of the Natural Logarithm

For any numbers $b > 0$ and $x > 0$, the natural logarithm satisfies the following rules:

1. *Product Rule*: $\ln bx = \ln b + \ln x$ — $\ln(ab) = \ln a + \ln b$
2. *Quotient Rule*: $\ln \frac{b}{x} = \ln b - \ln x$ — $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$
3. *Reciprocal Rule*: $\ln \frac{1}{x} = -\ln x$ — $\ln\left(\frac{1}{x}\right) = \ln 1 - \ln x = 0 - \ln x = \ln x$
4. *Power Rule*: $\ln x^r = r \ln x$. — $\ln(x^r) = r \ln x$.

Example.

(a) $\ln 4 + \ln \sin x = \ln(4 \sin x)$ — $\ln 4 + \ln \sin x = \ln(4 \sin x)$

(b) $\ln \frac{x+1}{2x-3} = \ln(x+1) - \ln(2x-3)$ — $\ln\left(\frac{x+1}{2x-3}\right) = \ln(x+1) - \ln(2x-3)$

(c) $\ln \frac{1}{8} =$ — $\ln\left(\frac{1}{8}\right) = \ln 1 - \ln 8 = -\ln(2^3) = -3 \ln 2$

The Graph and Range of $\ln x$

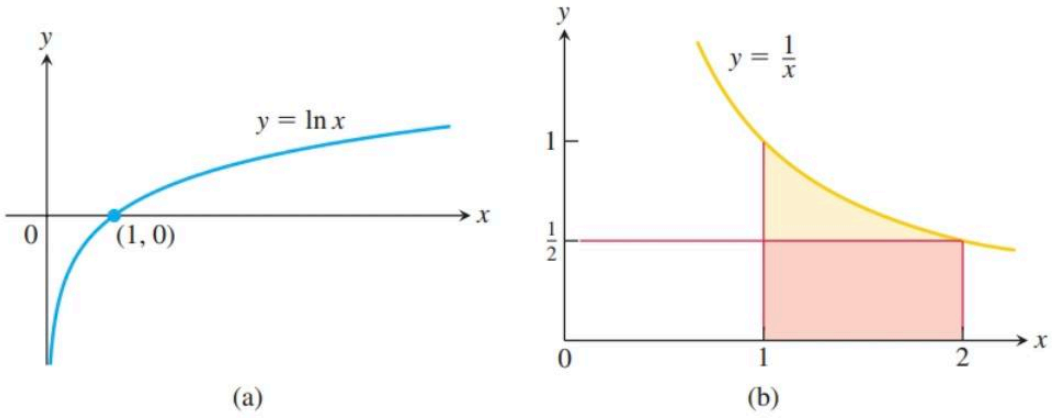


FIGURE 7.9 (a) The graph of the natural logarithm. (b) The rectangle of height $y = 1/2$ fits beneath the graph of $y = 1/x$ for the interval $1 \le x \le 2$.

The derivative $\frac{d}{dx}(\ln x) = \frac{1}{x}$ is positive for $x > 0$, so $\ln x$ is an increasing function of x.

The second derivative, $\frac{d^2}{dx^2}(\ln x) = -\frac{1}{x^2}$, is negative, so the graph of $\ln x$ is concave down.

We defined $\ln x$ for $x > 0$, so the domain of $\ln x$ is the set of positive real numbers.

The Integral $\int \frac{1}{x} dx$

$$\int \frac{1}{x} dx = \ln|x| + C.$$

If u is a differentiable function that is never zero, then

$$\int \frac{1}{u} du = \ln|u| + C.$$

Whenever $u = f(x)$ is a differentiable function that is never zero, we have $du = f'(x)dx$ and

$$\int f'(x)(f(x))^n dx = \frac{(f(x))^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C.$$

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C.$$

$\frac{d}{dx}(3 + 2\sin\theta) = 2\cos\theta$

Example. We rewrite an integral so that it has the form $\int \frac{1}{u} du$.

$$\int \frac{4\cos\theta}{3+2\sin\theta} d\theta \rightarrow \int \frac{f'(x)}{f(x)} dx$$

$$= 2\int \frac{2\cos\theta}{3+2\sin\theta} d\theta$$

$$= 2\ln|3+2\sin\theta| + C.$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{4\cos\theta}{3+2\sin\theta} d\theta$$

$$= 2\int_{-\pi/2}^{\pi/2} \frac{2\cos\theta\, d\theta}{3+2\sin\theta}$$

$$= 2\int_1^5 \frac{du}{u}$$

$$= 2\left[\ln|u|\right]_1^5 = 2(\ln 5 - \ln 1)$$

$$= 2\ln 5.$$

$u = 3 + 2\sin\theta$

$du = 2\cos\theta\, d\theta$

$\theta = -\pi/2 \quad u = 3 + 2(-1) = 1$

$\theta = \pi/2 \quad u = 3 + 2(1) = 5$

$\ln\left(\frac{1}{x}\right) = -\ln x$

The Integrals of $\tan x, \cot x, \sec x$, and $\operatorname{cosec} x$

$$\int \tan x\, dx = -\int \frac{-\sin x}{\cos x} dx = -\ln|\cos x| + C = \ln|(\cos x)^{-1}| + C$$

$$= \ln\left|\frac{1}{\cos x}\right| + C$$

$$= \ln|\sec x| + C.$$

$$\int \cot x\, dx = \int \frac{\cos x}{\sin x} dx = \ln|\sin x| + C.$$

$$\rightarrow \int \sec x\, dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx = \ln|\sec x + \tan x| + C$$

$$\rightarrow \int \operatorname{cosec} x\, dx = -\int \frac{-\operatorname{cosec} x(\operatorname{cosec} x + \cot x)}{\operatorname{cosec} x + \cot x} dx$$

$$= -\ln|\operatorname{cosec} x + \cot x| + C.$$

Integrals of the tangent, cotangent, secant, and cosecant functions

$$\int \tan x\,dx = \ln|\sec x| + C \qquad \int \sec x\,dx = \ln|\sec x + \tan x| + C$$

$$\int \cot x\,dx = \ln|\sin x| + C \qquad \int \csc x\,dx = -\ln|\csc x + \cot x| + C$$

$\frac{d}{dx}(\cos 2x) = -2\sin 2x$

Example.

$$\int_0^{\frac{\pi}{6}} \tan 2x\,dx = \int_0^{\frac{\pi}{6}} \frac{\sin 2x}{\cos 2x}\,dx$$

$$= -\frac{1}{2}\int_0^{\frac{\pi}{6}} \frac{-2\sin 2x}{\cos 2x}\,dx$$

$$= -\frac{1}{2}\left[\ln|\cos 2x|\right]_0^{\frac{\pi}{6}}$$

$$= -\frac{1}{2}\left(\ln\left|\cos\frac{\pi}{3}\right| - \ln|\cos 0|\right)$$

$$= -\frac{1}{2}\left(\ln\tfrac{1}{2} - \ln 1\right)$$

$$= -\frac{1}{2}\ln\tfrac{1}{2}$$

$$= \tfrac{1}{2}\ln 2.$$

$\ln(ab) = \ln a + \ln b$

$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$

$\ln(x^r) = r\ln x$

Find $\frac{dy}{dx}$ if

$$y = \frac{(x^2+1)(x+3)^{\frac{1}{2}}}{x-1}, \qquad x > 1.$$

$$\ln y = \ln\left(\frac{(x^2+1)(x+3)^{1/2}}{x-1}\right)$$

$$= \ln(x^2+1) + \ln(x+3)^{1/2} - \ln(x-1)$$

$$= \ln(x^2+1) + \tfrac{1}{2}\ln(x+3) - \ln(x-1).$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{2x}{x^2+1} + \frac{1}{2}\left(\frac{1}{x+3}\right) - \frac{1}{x-1}$$

$$\frac{dy}{dx} = y\left(\frac{2x}{x^2+1} + \frac{1}{2}\left(\frac{1}{x+3}\right) - \frac{1}{x-1}\right)$$

$$= \left(\frac{(x^2+1)(x+3)^{1/2}}{x-1}\right)\left(\frac{2x}{x^2+1} + \frac{1}{2}\left(\frac{1}{x+3}\right) - \frac{1}{x-1}\right).$$

Logarithmic Differentiation

The calculation of derivatives of complicated functions involving products, quotients, or powers can often be simplified by taking logarithms.

Method:

1. Take natural logarithms on both sides of equation $y = f(x)$, and simplify by using Logarithms Property.

2. Differentiate implicitly with respect to x.

3. Solve the equation for y'.

Example.

Differentiate $y = x^{\sqrt{x}}$.

Solution.

$$y = x^{\sqrt{x}}$$

$$\ln y = \ln x^{\sqrt{x}}$$

$$\ln y = \sqrt{x} \ln x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \sqrt{x}\left(\frac{1}{x}\right) + \ln x \left(\frac{1}{2\sqrt{x}}\right)$$

$$\frac{dy}{dx} = y\left(\sqrt{x}\left(\frac{1}{x}\right) + \ln x \left(\frac{1}{2\sqrt{x}}\right)\right)$$

$$\frac{dy}{dx} = x^{\sqrt{x}}\left(\frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}}\right)$$

$f \circ g$

$f^{-1}(f(x)) = x = f(f^{-1}(x))$

$\ln(e^x) = x = e^{\ln x}$

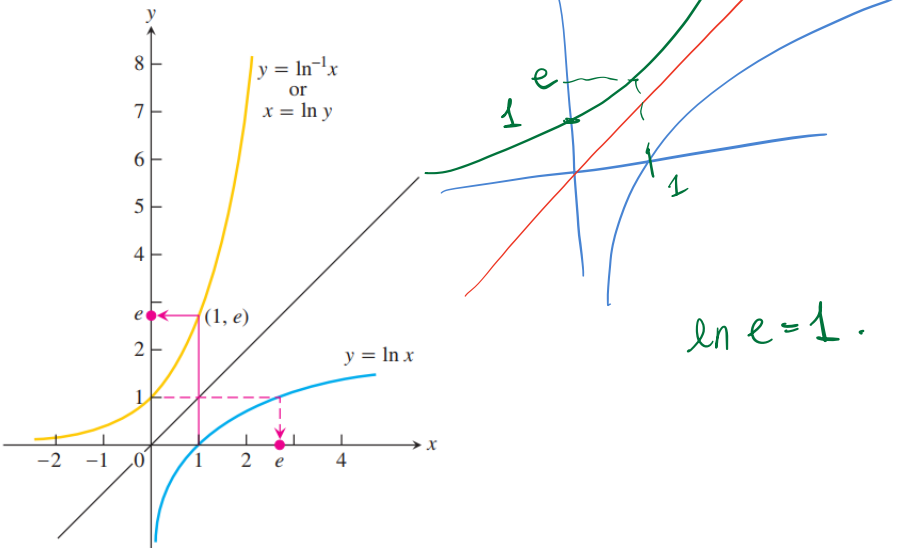
The Exponential Function

We introduce the exponential function, e^x as the inverse of $\ln x$.

e^x = inverse of $\ln x$.

The Inverse of $\ln x$ and the Number e

The function $\ln x$, being an increasing function of x with domain $(0, \infty)$ and range $(-\infty, \infty)$, has inverse $e^x = \ln^{-1} x$ with domain $(-\infty, \infty)$ and range $(0, \infty)$. The graph of e^x is the graph of $\ln x$ reflected across the line $y = x$.



$\ln e = 1$.

We note that e is not a rational number, later we will see one way to express it as a limit.

$$e = 2.718281828459045 \ldots$$

Definition. *The Natural Exponential Function*

For every real number x, $e^x = \ln^{-1} x = \exp(x)$.

For the first time we have a precise meaning for a number raised to an irrational power.

Inverse Equations for e^x and $\ln x$

$$e^{\ln x} = x, \quad (\text{all } x > 0)$$
$$\ln(e^x) = x, \quad (\text{all } x)$$

Example. Solve the equation $e^{2x-6} = 4$ for x.

$e^{2x-6} = 4$

$\ln e^{(2x-6)} = \ln 4$

$(2x-6)\ln e = \ln 2^2$

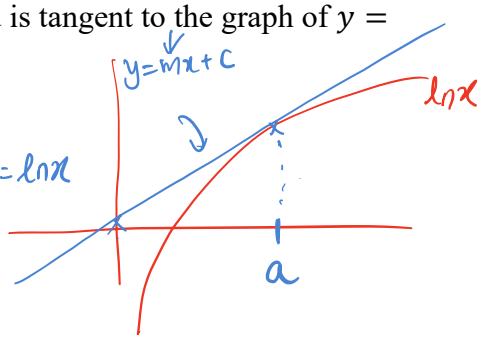
$(2x-6)\,1 = 2\ln 2$

$2x - 6 = 2\ln 2$

$2x = 2\ln 2 + 6$

$x = \ln 2 + 3$

Example. A line with slope m passes through the origin and is tangent to the graph of $y = \ln x$. What is the value of m?



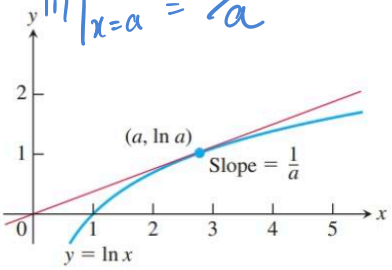


FIGURE 7.11 The tangent line intersects the curve at some point $(a, \ln a)$, where the slope of the curve is $1/a$ (Example 2).

The Number e Expressed as a Limit

Theorem. *The Number e as a Limit*

The number e can be calculated as the limit

$$e = \lim_{x \to 0} (1 + x)^{\frac{1}{x}}.$$

By substituting $y = 1/x$, we can also express the limit in the Theorem as

$$e = \lim_{y \to \infty} \left(1 + \frac{1}{y}\right)^{y}.$$

The Derivative of e^x

$$\frac{d}{dx}(e^x) = e^x$$

In particular, If u is any differentiable function of x, then

$$\frac{d}{dx}(e^u) = e^u \frac{du}{dx}.$$

Example. *Differentiating an Exponential*

$$\frac{d}{dx}(5e^x) = 5e^x$$

Example. *Applying the Chain Rule with Exponentials*.

(a) $\frac{d}{dx}e^{-x} = e^{-x}\frac{d}{dx}(-x) = -e^{-x}$

(b) $\frac{d}{dx}e^{\sin x} = e^{\sin x}(\cos x)$

(c) $\frac{d}{dx}\left(e^{\sqrt{3x+1}}\right) =$

(d) $\frac{d}{dx}(5e^{-x}) =$

$\frac{d}{dx}(5e^x) = 5\frac{d}{dx}e^x$

$= 5e^x$

$\frac{d}{dx}(e^{-x}) = e^{-x}\cdot\frac{d}{dx}(-x)$

$= -e^{-x}$

$\frac{d}{dx}(e^{\sin x}) = e^{\sin x}\cdot\cos x$

$\frac{d}{dx}(e^{\sqrt{3x+1}}) = e^{\sqrt{3x+1}}\cdot\left(\frac{3}{2\sqrt{3x+1}}\right)$.

$\frac{d}{dx}(5e^{-x}) = 5\frac{d}{dx}(e^{-x})$

$= -5e^{-x}$.

The Antiderivative of the Exponential Function

$$\int e^x\,dx = e^x + C.$$

$\frac{d}{dx}e^x = e^x$

$\int e^x\,dx = e^x + C$.

$\frac{d}{dx}(e^{3x}) = 3e^{3x}$

Example.

(a) $\int_0^{\ln 2} e^{3x}\,dx =$ $\frac{e^{3x}}{3}\Big]_0^{\ln 2}$ $= \frac{1}{3}\left[e^{3\ln 2} - e^0\right]$

$= \frac{1}{3}\left(e^{\ln 2^3} - 1\right)$ $= \frac{1}{3}(8-1)$

$= 7/3$.

(b) $\int_0^{\frac{\pi}{2}} e^{\sin x}\cos x\,dx =$

$= \left[e^{\sin x}\right]_0^{\frac{\pi}{2}} = e^{\sin \pi/2} - e^{\sin 0}$

$= e - 1$

The derivative of e^x exists and is everywhere positive, and that it is a continuous and increasing function. Since the second derivative of e^x is also e^x and everywhere positive, the graph is concave up.

$e^x > 0$

$$\lim_{x\to-\infty} e^x = 0 \qquad \text{and} \qquad \lim_{x\to+\infty} e^x = \infty.$$

From the first of these limits we see that the x-axis is a horizontal asymptote of the graph $y = e^x$.

Laws of Exponents

Even though e^x is defined in a seemingly roundabout way as $\ln^{-1} x$, it obeys the familiar laws of exponents from algebra.

Theorem. For all numbers x and y, the natural exponential e^x obeys the following laws:

1. $e^x e^y = e^{x+y}$

2. $e^{-x} = \frac{1}{e^x}$

3. $\frac{e^x}{e^y} = e^{x-y}$

4. $(e^x)^r = e^{rx}$, if r is rational.

$a = e^x \Rightarrow \ln a = x$
$b = e^y \Rightarrow \ln b = y$
$x + y = \ln a + \ln b$
$= \ln(ab)$
$e^{x+y} = ab = e^x \cdot e^y$

Proof of Law 1.

Proof of Law 4.

The General Exponential Function a^x

Definition

For any numbers $a > 0$ and x, the exponential function with base a is

$$a^x = e^{x \ln a}.$$

$a^x = e^{\ln(a^x)} = e^{x \ln a}$

Proof of the Power Rule (General Version)

The definition of the general exponential function enables us to make sense of raising any positive number to a real power n, rational or irrational. That is, we can define the power function $y = x^n$ for any exponent n.

Definition
For any $x > 0$ and any real number n.

$$x^n = e^{n \ln x}.$$

Because the logarithm and exponential functions are inverses of each other, the definition gives

$$\ln x^n = n \ln x, \text{ for all real numbers } n.$$

General Power Rule for Derivatives

For $x > 0$ and any real number n,

$$\frac{d}{dx} x^n = nx^{n-1}.$$

If $x \leq 0$, then the formula holds whenever the derivative, x^n, and x^{n-1} all exist.

The Derivative of a^x

To find this derivative, we start with the equation $a^x = e^{x \ln a}$, which defines a^x for any positive a. Then we have

$$\frac{d}{dx} a^x = \frac{d}{dx} e^{x \ln a}$$

$$= e^{x \ln a} \frac{d}{dx}(x \ln a)$$

$$= a^x \ln a.$$

$y = a^x$
$\ln y = \ln a^x$
$= x \ln a$
$\frac{1}{y}\frac{dy}{dx} = \ln a$
$\frac{dy}{dx} = y \ln a$
$= a^x \ln a$

$$\frac{d}{dx} a^x = a^x \ln a.$$

$\frac{d}{dx}(a^x) = a^x \ln a$

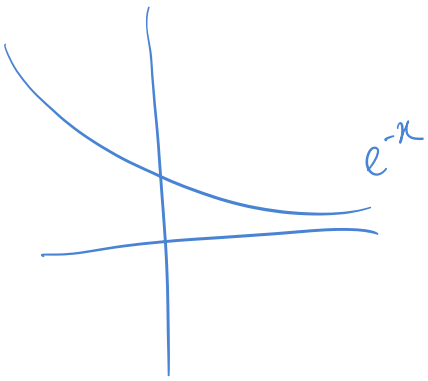
If $a = e$, then $\ln a = 1$ and the derivative of a^x simplifies to

$$\frac{d}{dx} a^u = a^u \ln a \frac{du}{dx}.$$

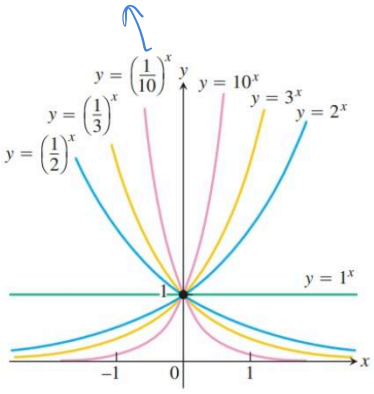
$\frac{d}{dx}(a^{3x+1})$
$y = a^{3x+1}$
$\ln y = \ln(a^{3x+1})$
$= (3x+1)\ln a$
$\frac{1}{y}\frac{dy}{dx} = 3 \ln a$
$\frac{dy}{dx} = 3(\ln a)\, a^{3x+1}$

$\frac{d}{dx} a^x = a^x \ln a$ $\quad \int a^x dx = \frac{a^x}{\ln a} + C$

$\int a^x dx = \frac{1}{\ln a} a^x + C$.



$\left(\frac{1}{10}\right)^x = 10^{-x}$



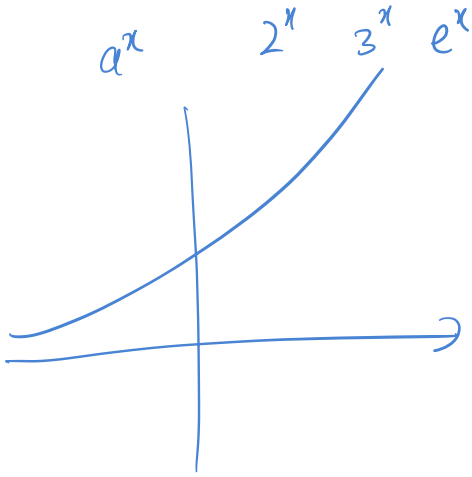


FIGURE 7.13 Exponential functions decrease if $0 < a < 1$ and increase if $a > 1$. As $x \to \infty$, we have $a^x \to 0$ if $0 < a < 1$ and $a^x \to \infty$ if $a > 1$. As $x \to -\infty$, we have $a^x \to \infty$ if $0 < a < 1$ and $a^x \to 0$ if $a > 1$.

Example. We find the derivatives and integrals involving general exponential functions.

(a) $\frac{d}{dx}\, 3^x =$ $3^x \ln 3$

$y = 3^x$

$\ln y = x \ln 3$

$\frac{1}{y}\frac{dy}{dx} = \ln 3 \Rightarrow \frac{dy}{dx} = 3^x \ln 3.$

(b) $\frac{d}{dx}\, 3^{-x} =$ $3^{-x} \ln 3\,(-1) = (-\ln 3)\, 3^{-x}$.

$y = 3^{\sin x}$

$\ln y = \sin x (\ln 3)$

$\frac{1}{y}\frac{dy}{dx} = (\ln 3)\cos x$

$\frac{dy}{dx} = 3^{\sin x}(\ln 3)\cos x$

(c) $\frac{d}{dx}\, 3^{\sin x} =$ $3^{\sin x} \ln 3 \cdot (\cos x)$

(d) $\int 2^x\, dx =$ $\frac{2^x}{\ln 2} + C$

(e) $\int 2^{\sin x} \cos x\, dx =$ $\frac{2^{\sin x}}{\ln 2} + C$.

$\frac{d}{dx}\, 2^{\sin x} = 2^{\sin x} \ln 2 \cdot \cos x$

Example. Differentiate $f(x) = x^x$, for $x > 0$.

$\ln x = \log_e x$

$\log_{10} x$

$\int \ln x\, dx = ?$

$\frac{d}{dx} \ln x = \frac{1}{x}$

$\frac{d}{dx} e^x = e^x$

$\int \frac{1}{x}\, dx = \ln|x| + C$

$\int e^x dx = e^x + C$

$u = f(x)$ $\frac{d}{dx} \ln u = \frac{1}{u}\frac{du}{dx}$

$\frac{d}{dx} e^u = e^u \frac{du}{dx}$

OCC 1

2 5 12 3 4 5

In words | two Five one two three four five.

Logarithms with Base a

If a is any positive number other than 1, the function a^x is one-to-one and has a nonzero derivative at every point.

Definition.
For any positive number $a \neq 1$, the logarithm of x with base a, denoted by $\log_a x$, is the inverse function of a^x.

$\log_2 3 = \frac{\log_{10} 3}{\log_{10} 2}$

Inverse Equations for a^x and $\log_a x$.

$$a^{\log_a x} = x \qquad (x > 0)$$
$$\log_a(a^x) = x \qquad (\text{all } x)$$

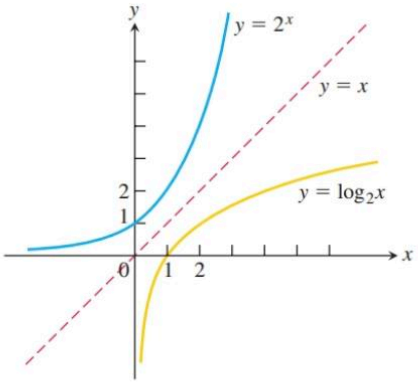


FIGURE 7.14 The graph of 2^x and its inverse, $\log_2 x$.

$\frac{d}{dx}(\log_a x)$

$= \frac{d}{dx}\left(\frac{\log_e x}{\log_e 2}\right)$

$= \frac{d}{dx}\left(\frac{\ln x}{\ln 2}\right)$

$= \frac{1}{\ln 2}\left(\frac{d}{dx}\ln x\right)$

$= \frac{1}{\ln 2}\left(\frac{1}{x}\right)$

$\frac{d}{dx}(2^x)$

$y = 2^x$

$\ln y = \ln 2^x$

$\ln y = x \ln 2.$

$\frac{1}{y}\frac{dy}{dx} = \ln 2 \Rightarrow \frac{dy}{dx} = y \ln 2$

$\frac{d}{dx}(2^x) = 2^x \ln 2$

$\frac{d}{dx}(a^x) = a^x \ln a.$

Derivatives and Integrals involving $\log_a x$

To find derivatives or integrals involving base a logarithms, we can convert them into natural logarithms.

$$\frac{d}{dx}(\log_a x) = \frac{d}{dx}\left(\frac{\ln x}{\ln a}\right) = \frac{1}{\ln a}\frac{d}{dx}(\ln x) = \frac{1}{\ln a}\left(\frac{1}{x}\right).$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

In particular,

$$\frac{d}{dx}(\log_a u) = \frac{1}{\ln a}\left(\frac{1}{u}\right)\frac{du}{dx}.$$

$\frac{d}{dx}\log_a f(x)$

$= \frac{d}{dx}\left(\frac{\ln f(x)}{\ln a}\right)$

$= \frac{1}{\ln a}\frac{d}{dx}(\ln f(x))$

$= \frac{1}{\ln a}\left(\frac{1}{f(x)}\right)f'(x)$

$= \frac{1}{\ln a}\left(\frac{f'(x)}{f(x)}\right).$

Example. We illustrate the derivative and integral results.

(a) $\frac{d}{dx}\log_{10}(3x+1) =$

(b) $\int \frac{\log_2 x}{x}\,dx =$

Note. ***The Derivative of a^u, where a is any positive real number.***

If $a > 0$ and u is a differentiable function of x, then a^u is a differentiable function of x and

$$\frac{d}{dx}(a^u) = a^u \ln a \frac{du}{dx}$$

In particular,

$$\frac{d}{dx}(a^x) = a^x \ln a$$

Proof.

Let $y = a^u$. Then $\ln y = u \ln a$.

$$\frac{1}{y}\cdot\frac{dy}{dx} = \ln a \frac{du}{dx}$$

$$\frac{dy}{dx} = a^u \ln a \frac{du}{dx}$$

In particular, if $u = x$, then $\frac{du}{dx} = 1$.

$$\frac{dy}{dx} = a^x \ln a.$$

Indeterminate Forms and L'Hopital's Rule.

L'Hôpital

The rule is known today as l'Hôpital's Rule, after Guillaume de l'Hôpital. He was a French nobleman who wrote the first introductory differential calculus text, where the rule first appeared in print.

Indeterminate Form 0/0

$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}$

Theorem. *L'Hopital's Rule (First Form)*
Suppose that $f(a) = g(a) = 0$, that $f'(a)$ and $g'(a)$ exist, and that $g'(a) \neq 0$. Then

$0/0$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

Proof.

$$\frac{f'(a)}{g'(a)} = \frac{\lim_{x\to a}\frac{f(x)-f(a)}{x-a}}{\lim_{x\to a}\frac{g(x)-g(a)}{x-a}} = \lim_{x\to a}\frac{\frac{f(x)-f(a)}{x-a}}{\frac{g(x)-g(a)}{x-a}} = \lim_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)} = \lim_{x\to a}\frac{f(x)-0}{g(x)-0} = \lim_{x\to a}\frac{f(x)}{g(x)}.$$

Theorem. *L'Hopital's Rule*
Suppose that $f(a) = g(a) = 0$, that $f'(a)$ and $g'(a)$ exist, and that $g'(a) \neq 0$. Then

$f(a) = g(a) = 0$
$f'(a)$, $g'(a)$ exist
$g'(a) \neq 0$

$0/0$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

Example. *Using L'Hopital's Rule*

(a) $\lim_{x\to 0}\frac{3x-\sin x}{x} = \frac{3-\cos x}{1}|_{x=0} = 2.$

$\lim_{x\to 0}\frac{3x-\sin x}{x}$ → $0/0$
$= \lim_{x\to 0}\frac{3-\cos x}{1}$ → $2/1$
$= \frac{2}{1} = 2.$

$\lim_{x\to 0}\frac{3x}{x} - \frac{\sin x}{x}$
$= \lim_{x\to 0}\left(3 - \frac{\sin x}{x}\right)$, $\frac{\sin x}{x} \to 1$
$= 2$

(b) $\lim_{x\to 0}\frac{\sqrt{1+x}-1}{x} = \frac{\frac{1}{2\sqrt{1+x}}}{1}|_{x=0} = \frac{1}{2}.$

$\lim_{x\to 0}\left(\frac{\sqrt{1+x}-1}{x}\right)$, $1-1=0$ → $0/0$
$= \lim_{x\to 0}\frac{\frac{1}{2\sqrt{1+x}}}{1}$, $\frac{1}{2}$, $\neq 0$
$= 1/2.$

(c) $\lim_{x\to 0}\frac{\sqrt{1+x}-1-\frac{x}{2}}{x^2} =$

$\lim_{x\to 0}\frac{\sqrt{1+x}-1-\frac{x}{2}}{x^2}$ → $0/0$
$= \lim_{x\to 0}\frac{\frac{1}{2\sqrt{1+x}}-\frac{1}{2}}{2x}$ → $0/0$
$= \lim_{x\to 0}\frac{\frac{-1}{4(1+x)^{3/2}}}{2}$, $\frac{-1}{4}$, 2
$= -1/8$

(d) $\lim_{x\to 0}\frac{x-\sin x}{x^3} =$

$\lim_{x\to 0}\frac{x-\sin x}{x^3}$ → $0/0$
$= \lim_{x\to 0}\frac{1-\cos x}{3x^2}$ → $0/0$
$= \lim_{x\to 0}\frac{-(-\sin x)}{6x}$
$= \lim_{x\to 0}\frac{1}{6}\left(\frac{\sin x}{x}\right)$, $\frac{\sin x}{x} \to 1$
$= \frac{1}{6}\lim_{x\to 0}\frac{\cos x}{1}$, $\cos x \to 1$, $1 \neq 0$
$= \frac{1}{6}(1) = 1/6.$

(e) $\lim_{x\to\infty}\frac{\ln\left(1+\frac{1}{x}\right)}{\sin\left(\frac{1}{x}\right)} =$

$\lim_{x\to\infty}\frac{\ln(1+1/x)}{\sin(1/x)}$, $\ln 1 = 0$, $\sin 0 = 0$
$= \lim_{x\to\infty}\frac{\frac{1}{1+1/x}\left(-\frac{1}{x^2}\right)}{\cos(1/x)\left(-\frac{1}{x^2}\right)}$
$= \lim_{x\to\infty}\frac{\frac{1}{1+1/x}}{\cos(1/x)}$, $\frac{1}{1+1/x} \to 1$, $\cos(1/x) \to 1$
$= 1.$

$0/0$, ∞/∞, $0/\infty$

Using L'Hopital Rule.
To find

$$\lim_{x\to a}\frac{f(x)}{g(x)}$$

by L'Hopital's Rule, continue to differentiate f and g, as long as we still get the form $0/0$ at $x = a$. But as soon as one or the other of these derivatives is different from zero at $x = a$ we stop differentiating. L'Hopital Rule does not apply when either the numerator or denominator has a finite nonzero limit.

Example.

$$\lim_{x\to 0}\frac{1-\cos x}{x+x^2} =$$

$1-1=0$, 0, by L'Hôpital rule. $\frac{0}{0}$

$$= \lim_{x\to 0}\frac{\sin x}{1+2x}$$

0, $1+2x \to 1 \neq 0$, cannot apply L'Hôpital rule.

$$= 0$$

L'Hopital's Rule applies to one-sided limits as well, which is apparent from the proof of *Theorem of L'Hopital's Form*.

Example. *Using L'Hopital Rule with One-Sided Limits.*

(a) $\lim_{x\to 0^+}\frac{\sin x}{x^2} = \lim_{x\to 0^+}\frac{\cos x}{2x} = \infty$. (positive for $x > 0$)

(b) $\lim_{x\to 0^-}\frac{\sin x}{x^2} = \lim_{x\to 0^-}\frac{\cos x}{2x} = -\infty$. (positive for $x < 0$)

$\lim_{x\to 0^+}\frac{\sin x}{x^2}$: $\sin x \to 0$, $x^2 \to 0$ (+ve); $= \lim_{x\to 0^+}\frac{\cos x}{2x}$: $\cos x \to 1$, $2x \to 0$ (+ve); $= +\infty$

$\lim_{x\to 0^-}\frac{\sin x}{x^2} = \lim_{x\to 0^-}\frac{\cos x}{2x} = -\infty$

Indeterminate Forms ∞/∞, $\infty \cdot 0$, $\infty - \infty$

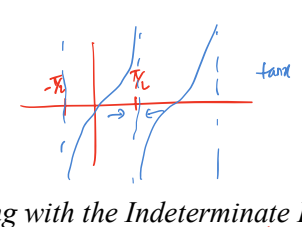
If $f(x) \to \pm\infty$ and $g(x) \to \pm\infty$ as $x \to a$, then

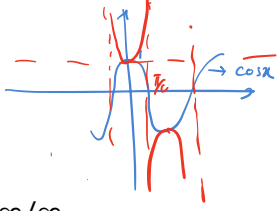
$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$$

provided the limit on the right exists.

Note. In the notation $x \to a$, a may be either finite or infinite. Moreover, $x \to a$ may be replaced by the one-sided limits $x \to a^+$ or $x \to a^-$.

$\frac{1}{\cos x} = \sec x$





Example. *Working with the Indeterminate Form* ∞/∞

Find

(a) $\lim_{x\to\frac{\pi}{2}} \frac{\sec x}{1+\tan x}$

(b) $\lim_{x\to\infty} \frac{x+2x^2}{3x^2+5x}$

(c) $\lim_{x\to\infty} \frac{\ln x}{2\sqrt{x}} =$

(d) $\lim_{x\to\infty} \frac{e^x}{x^2} =$

Solution.

(a) The numerator and denominator are discontinuous at $x = \frac{\pi}{2}$, so we investigate the one-sided limits there. To apply L'Hopital Rule, we can choose I to be open interval with $x = \frac{\pi}{2}$ as an endpoint.

$$\lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{\sec x}{1+\tan x} = \lim_{x\to\left(\frac{\pi}{2}\right)^-} \frac{\sec x \tan x}{\sec^2 x} = \lim_{x\to\left(\frac{\pi}{2}\right)^-} \sin x = 1.$$

The right-hand limit is 1 also, with $-\infty/-\infty$ as the indeterminate form. Therefore, the two-sided is equal to 1.

(b)

$$\lim_{x\to\infty} \frac{x+2x^2}{3x^2+5x} = \lim_{x\to\infty} \frac{1+4x}{6x+5} = \lim_{x\to\infty} \frac{4}{6} = \frac{2}{3}.$$

Example. *Working with the Indeterminate Form* $\infty \cdot 0$

(a) Find $\lim_{x\to\infty} \left(x \sin\frac{1}{x}\right)$.

Solution. Let $h = 1/x$.

$$\lim_{x\to\infty} \left(x \sin\frac{1}{x}\right) = \lim_{x\to\infty} \frac{\left(\sin\frac{1}{x}\right)}{\frac{1}{x}} = \lim_{h\to 0^+} \frac{\sin h}{h} = 1.$$

(b) Find $\lim_{x\to 0^+} \left(\sqrt{x}\ln x\right)$.

Example. *Working with the Indeterminate Form* $\infty - \infty$

Find $\lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{1}{x}\right)$.

Solution. If $x \to 0^+$, then $\sin x \to 0^+$ and

$$\frac{1}{\sin x}-\frac{1}{x} \to \infty - \infty.$$

Similarly, if $x \to 0^-$, then $\sin x \to 0^-$ and

$$\frac{1}{\sin x}-\frac{1}{x} \to -\infty - (-\infty) = -\infty + \infty.$$

Neither form reveals what happens in the limit. To find out, we first combine the fractions:

$$\frac{1}{\sin x}-\frac{1}{x} = \frac{x - \sin x}{x \sin x}.$$

Then apply L'Hopital Rule to the result:

$$\lim_{x\to 0}\left(\frac{1}{\sin x}-\frac{1}{x}\right) = \lim_{x\to 0}\frac{x-\sin x}{x\sin x} = \lim_{x\to 0}\frac{1-\cos x}{\sin x + x\cos x} = \lim_{x\to 0}\frac{\sin x}{2\cos x - x\sin x} = \frac{0}{2} = 0.$$

Indeterminate Powers

Limits that lead to the indeterminate form 1^∞, 0^0, and ∞^0.

If $\lim_{x\to a}(\ln f(x)) = L$, then

$$\lim_{x\to a} f(x) = \lim_{x\to a} e^{\ln f(x)} = e^L.$$

Here a may be either finite or infinite.

Example. Apply L'Hopital's Rule to show that $\lim_{x\to 0^+}(1+x)^{\frac{1}{x}} = e$.

Example. Find $\lim_{x\to\infty} x^{\frac{1}{x}}$.

Theorem *Cauchy's Mean Value Theorem*

Suppose functions f and g are continuous on $[a, b]$ and differentiable thoughout (a, b) and also suppose $g'(x) \neq 0$ throughout (a, b). Then there exists a number $c \in (a, b)$ at which

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

$\exists c \in (a,b)$ s.t.

$$\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}.$$

$g(x) = x$ continuous $[a, b]$ diff. (a, b)

$\exists c \in (a,b)$

$$\frac{f'(c)}{1} = \frac{f(b)-f(a)}{b-a}$$

$(g(x)=x)$.

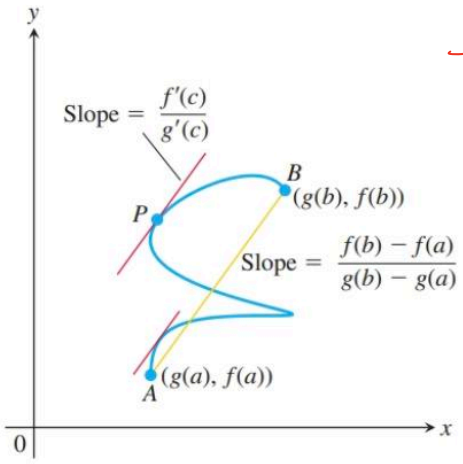


FIGURE 7.20 There is at least one point P on the curve C for which the slope of the tangent line to the curve at P is the same as the slope of the secant line joining the points $A(g(a), f(a))$ and $B(g(b), f(b))$.